Water Ledger Ohio: methods brief

Model version 1.2 · data snapshot 2026-10-29 · waterledger.org · github.com/northlinehydrology/waterledger

What it estimates

For any city, town, village, township or county in Ohio, Water Ledger estimates how much water a data center campus of a given size and cooling design would take, use up and send back, on an average day and on a 1-in-10-year peak day, and sets that beside the public water system, the sewer plant, the streams downstream and the electric utility that would serve it. It uses public data only and is a screening estimate, not a design.

Data

Category	Source	In Ohio
Places	Census TIGERweb: incorporated places; county subdivisions that are governments; counties	2,318 places: 922 cities, towns and villages; 1,308 townships; 88 counties
Water supply	EPA community water system service areas, with SDWIS population and connections	1,077 systems; 76% of places have one mapped
Wastewater	EPA Clean Watersheds Needs Survey 2022 (design flow) matched to ECHO (recent flow, receiving water)	669 plants; 100% of places have one within 40 km
Receiving waters	USGS Watershed Boundary Dataset; EPA ATTAINS (21OHIO, 2024 report)	1,198 watersheds with a downstream path; 1,419 watersheds on those paths
Climate	NOAA ISD station records, 1995-2024, one station per NOAA climate division; Atlas 14 design storms; Climate at a Glance rainfall	10 climate divisions
Energy	HIFLD retail service territories (PNNL copy); EIA-861 sales and peaks; Siddik et al. 2024 water per kWh by grid region	121 utilities with loads; grid regions: LGEE, MISO, PJM

How an estimate is made

Cooling water. Hour by hour over 30 years of weather, the campus's heat load is turned into evaporation, drift and blowdown for each cooling design: cooling towers, hybrid (evaporative only in warm weather), closed loop with evaporative assist on the hottest hours, and closed loop with no water for cooling. The evaporative share follows Rutberg et al. (2011); blowdown follows the DOE FEMP water balance.

Uncertainty. The model runs 200 times with its 15 cooling parameters drawn from published or estimated ranges (marked [A] where they stand in for a facility's own data). Results are the 10th, 50th and 90th percentiles; the peak day is the 90th percentile of each year's highest day.

Staff water and loop top-ups are added at the campus's size (people on site, and small leaks from closed loops).

Source and wastewater. The campus uses the public water system mapped in the place; with none mapped, a closed loop uses the nearest system serving over 3,300 people if it is within 10 km, and anything else uses new wells. Wastewater goes to the sewer when a plant is in reach and, for evaporative designs, has room for the blowdown; otherwise a septic system (closed loop) or the campus's own discharge.

Downstream. Each place's main watershed is followed downstream (up to 25 watersheds, across state lines), with the impairments and cleanup plans (TMDLs) EPA lists for each.

How this differs from Michigan

Weather. Ohio uses NOAA station records (Findlay Airport (2 divisions); Cleveland-Hopkins International Ap; J.M.Cox Dayton International Airport; Port Columbus International Airport; Akron-Canton Regional Airport (2 divisions); Cina Muni Apt/Lukn Fd Apt; Tri-State/M.J.Fergus; Zanesville Municipal Arpt). Michigan uses ERA5 reanalysis. For cooling towers the two agree within about 1% in Michigan, but for hybrid and assist, which use water only on the hottest hours, station records give higher averages (4-50% higher in southern Michigan, 2-5 times at Traverse City and Alpena), so hybrid and assist figures are not strictly comparable across the two.

Not yet included here: state-reported water use by sector, household wells, coldwater streams, lake-source cooling and state rules ("state rules not yet reviewed"). Utilities are listed by customers served, since the national territory map draws many small utilities as whole counties.

Example: 300 MW of IT capacity in Columbus

Million gallons a day: median, with the 10th to 90th percentile range in brackets. Peak day is the 1-in-10-year highest day. "Used up" is water evaporated or otherwise not returned. Defaults for water source and wastewater are filled in by the rules above (muni: public system; sewer: the nearest plant).

Design	Withdrawn, average day	Withdrawn, peak day	Used up, average day	Water from / to
Cooling towers	2.40 (1.97-2.77)	2.96 (2.41-3.43)	1.92 (1.60-2.23)	muni / sewer
Hybrid	0.19 (0.12-0.28)	1.99 (1.63-2.48)	0.15 (0.09-0.22)	muni / sewer
Closed loop with assist	0.01 (0.01-0.03)	0.32 (0.19-0.47)	0.01 (<0.01-0.02)	muni / sewer
Closed loop	<0.01	<0.01	<0.01	muni / sewer

Checks on this snapshot

Coverage of the state's places: water system 76%, watersheds 100%, utilities 100%, climate division 100%, a sewer plant within 40 km 100%. 50 places chosen at random ran through every step and cooling design with 0 errors. Places with no mapped water system are mostly rural areas on private wells.

Limits

These are screening estimates from public data, for comparing designs and sizes before an applicant's numbers exist. They do not replace a facility's design data, a utility's capacity analysis or a permit review. The watershed path is followed watershed to watershed, not stream by stream. EPA assessments reflect each state's latest report. Weather is the past 30 years.

Planning-level hydrologic assessment. Not engineering design. Built with the open waterledger Python packages (waterledger 2.0.0); methods, parameters and sources: waterledger.org and the package repository. Northline Hydrology.